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2014.0 RANGE ROVER (LG), 303-05

ACCESSORY DRIVE - V8 N/A 5.0L PETROL/V8 S/C 5.0L PETROL

DIAGNOSIS AND TESTING

PRINCIPLES OF OPERATION

For a detailed description of the accessory drive belts and operation, refer to the relevant Description and Operation section in the Workshop Manual.

REFER to: Accessory Drive (303-05 Accessory Drive - V8 5.0L Petrol/V8 S/C 5.0L Petrol, Description and Operation).

INSPECTION AND VERIFICATION



CAUTION:

Diagnosis by substitution from a donor vehicle is NOT acceptable. Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle.

1. Verify the customer concern.

1. Visually inspect for obvious signs of mechanical damage.

Visual Inspection

MECHANICAL

- Drive belt condition (cracking/damage/contamination)
- Idler assembly
- Generator
- Engine cooling fan
- Tensioner assembly
- Engine coolant pump
- Air conditioning (A/C) compressor
- Torsional vibration damper
- Dynamic response pump
- Tensioner assembly
- Accessory drive belt

1. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step.
1. If the cause is not visually evident, verify the symptom and refer to the Symptom Chart, alternatively check for Diagnostic Trouble Codes (DTCs) and refer to the DTC Index.

SYMPTOM CHART



CAUTION:

If the engine is run without the accessory drive belts connected to eliminate driven components, diagnostic trouble codes, (DTCs) may be set which must be cleared before the vehicle is returned to the owner. The engine should not be run for more than 2-3 minutes with the belts disconnected. Failure to follow this instruction may result in damage to the vehicle.

SYMPTOM	POSSIBLE CAUSES	ACTION
Noise	▪ Belt condition ▪ Belt tension ▪ Pulleys misaligned ▪ Driven components (including tensioners)	▪ Check the belt condition (see visual inspection). Check the tensioner function. Check the pulley alignment. Check the driven components for excessive resistance to rotation. Rectify as necessary. ▪ Using the manufacturer approved diagnostic system, perform routine - Noise, vibration and harshness sensor diagnostic test - Accessory drive belt.
Drive belt does not hold tension	▪ Belt condition ▪ Tensioner fault	▪ Check the belt condition (see visual inspection). Check the tensioner function. Rectify as necessary.

DTC INDEX

For a list of Diagnostic Trouble Codes (DTCs) that could be logged on this vehicle, please refer to Section 100-00.

REFER to: Diagnostic Trouble Code (DTC) Index - V8 S/C 5.0L Petrol, DTC: Engine Control Module (ECM) (100-00 General Information, Description and Operation).

